1:-

```c
#include <stdio.h>

int main(){
    char letter;
    int number;
    float decimal;
    double bigvalue;

    scanf("%c", &letter);
    scanf("%d", &number);
    scanf("%f", &decimal);
    scanf("%lf", &bigvalue);

    printf("%c  is the letter\n", letter);
    printf("%d  is the number\n", number);
    printf("%f  is the decimal\n", decimal);
    printf("%lf is the bigvalue\n", bigvalue);

    return 0;
}
```

Output :-

```
A
14
3.14
3.1456797342868314868
A   is the letter
14   is the number
3.140000   is the decimal
3.145680 is the bigvalue
```

2:-

```c
#include <stdio.h>
int main() {
    float num1, num2, num3;
    printf("enter the values :");
    scanf("%f %f %f", &num1, &num2, &num3);

    int sum = num1 + num2 + num3;
    float average = sum / 3;

    printf("%d is the sum \n", sum);
    printf("%fis the average\n", average);

    return 0;
}
```

Output :-

```
enter the values :3 4 5
12 is the sum
4.000000is the average
```

3:-

```c
#include <stdio.h>
int main() {
    float fahrenheit, celsius;
    int choice;
    printf("1. Fahrenheit to Celsius\n");
    printf("2. Celsius to Fahrenheit\n");
    printf("Selection:(1/2) ");
    scanf("%d", &choice);
    if (choice == 1) {
        printf("Enter temperature in Fahrenheit: ");
        scanf("%f", &fahrenheit);
        celsius = (fahrenheit - 32) * 5 / 9;
        printf("Temperature in Celsius: %f\n", celsius);
    }
    else if (choice == 2) {
        printf("Enter temperature in Celsius: ");
        scanf("%f", &celsius);
        fahrenheit = (celsius * 9 / 5) + 32;
        printf("Temperature in Fahrenheit: %f\n", fahrenheit);
    }
    else {
        printf("Invalid choice.\n");
    }
    return 0;
}
```

Output:-

```
1. Fahrenheit to Celsius
2. Celsius to Fahrenheit
Selection:(1/2) 1
Enter temperature in Fahrenheit: 98
Temperature in Celsius: 36.666668
```

```
1. Fahrenheit to Celsius
2. Celsius to Fahrenheit
Selection:(1/2) 2
Enter temperature in Celsius: 38
Temperature in Fahrenheit: 100.400002
```

4:-

```c
#include <stdio.h>

int main() {
    const int  Rate = 5;
    float principal;
    int time;

    printf("Enter the principal amount: ");
    scanf("%f", &principal);

    printf("Enter the time in years: ");
    scanf("%d", &time);

    float simpleInterest = (principal * Rate * time) / 100;

    printf("Principal: %f\n", principal);
    printf("Rate of Interest: %d\n", Rate);
    printf("Time: %d years\n", time);
    printf("Simple Interest: %f\n", simpleInterest);

    return 0;
}
```

Output :-

```
Enter the principal amount: 10000
Enter the time in years: 5
Principal: 10000.000000
Rate of Interest: 5
Time: 5 years
Simple Interest: 2500.000000
```

5:-

```c
#include <stdio.h>
int main() {
    int A = 10, B = 5, C = 2, D = 4, E = 3, F = 6, G = 1;
    int i = 5;
    int result_1, result_2, result_3;
    //q1
    result_1 = A + B * C + (D * E) + F * G;
    //q2
    result_2 = A / B * C - B + A * D / 3;
    //q3
    int tempA = A, tempB = B;
    tempA += tempB *= C + 3;
    result_3 = tempA;
    //q4
    int J = (i++) + (++i);

    printf("Result a: %d\n", result_1);
    printf("Result b: %d\n", result_2);
    printf("Result c: %d\n", result_3);
    printf("Result d: %d\n", J);

    return 0;
}
```

Output:-

```
Result q1:  38
Result q2:  12
Result q3:  35
Result q4:  12
```

6:-

```c
#include <stdio.h>

int main() {
    int a, b, c;

    printf("Enter three numbers: ");
    scanf("%d %d %d", &a, &b, &c);

    int max_value = (a > b) ? ((a > c) ? a : c) : ((b > c) ? b : c);

    printf("The maximum number is: %d\n", max_value);

    return 0;
}
```

Output:-

```
Enter three numbers: 4 5 6
The maximum number is: 6
```

7:-

```c
#include <stdio.h>
int main() {
    int n1, n2, n3, n4;
    int max, min;
    printf("Enter four numbers: ");
    scanf("%d %d %d %d", &n1, &n2, &n3, &n4);
    max = n1;
    min = n1;
    if (n2 > max) {
        max = n2;
    } else if (n2 < min) {
        min = n2;
    }
    if (n3 > max) {
        max = n3;
    } else if (n3 < min) {
        min = n3;
    }
    if (n4 > max) {
        max = n4;
    } else if (n4 < min) {
        min = n4;
    }

    printf("Maximum: %d\n", max);
    printf("Minimum: %d\n", min);

    return 0;
}
```

Output:-

```
Enter four numbers: 4 5 6 9
Maximum: 9
Minimum: 4
```

8:-

```c
#include <stdio.h>
int main() {
    float initialReading, finalReading, units, totalBill;

    printf("Enter initial meter reading: ");
    scanf("%f", &initialReading);
    printf("Enter final meter reading: ");
    scanf("%f", &finalReading);

    units = finalReading - initialReading;
    if (units > 400) {
        totalBill = units * 6.5;
    } else if (units >= 300) {
        totalBill = units * 5.5;
    } else if (units >= 200) {
        totalBill = units * 4.5;
    } else if (units >= 100) {
        totalBill = units * 3.5;
    } else if (units >= 50) {
        totalBill = units * 2.5;
    } else {
        totalBill = units * 2;
    }

    printf("Total units consumed: %.2f\n", units);
    printf("Total electricity bill: ₹%.2f\n", totalBill);

    return 0;
}
```

Output:-

```
Enter initial meter reading: 456.98
Enter final meter reading: 999.99
Total units consumed: 543.01
Total electricity bill: ₹3529.56
```

9:-

```c
#include <stdio.h>
int main() {
    int choice;
    float num1, num2, result;
    printf("1. Add\n2. Subtract\n3. Multiply\n4. Divide\n");
    printf("Pick an option: ");
    scanf("%d", &choice);
    printf("Enter two numbers: ");
    scanf("%f %f", &num1, &num2);
    switch (choice) {
        case 1:
            result = num1 + num2;
            printf("Result: %f\n", result);
            break;
        case 2:
            result = num1 - num2;
            printf("Result: %f\n", result);
            break;
        case 3:
            result = num1 * num2;
            printf("Result: %f\n", result);
            break;
        case 4:
            if (num2 != 0){
                result = num1 / num2;
                printf("Result: %f\n", result);} else {
                printf("Cannot divide by zero\n");}
            break;
        default:
            printf("Invalid choice\n");
    }return 0;
```

Output:-

```
1. Add
2. Subtract
3. Multiply
4. Divide
Pick an option: 3
Enter two numbers: 4 5
Result: 20.000000
```

10:-

```c
#include <stdio.h>
int main() {
    int num1, num2, num3;
    printf("Enter three numbers: ");
    scanf("%d %d %d", &num1, &num2, &num3);

    if (num1 >= num2) {
        if (num1 >= num3) {
            printf("The largest number is: %d\n", num1);
        } else {
            printf("The largest number is: %d\n", num3);
        }
    } else {
        if (num2 >= num3) {
            printf("The largest number is: %d\n", num2);
        } else {
            printf("The largest number is: %d\n", num3);
        }
    }

    return 0;
}
```

Output:-

```
Enter three numbers: 2 3 4
The largest number is: 4
```

11:-

```c
#include <stdio.h>
int main() {
    int n, i, isPrime = 1;
    printf("Enter a number: ");
    scanf("%d", &n);

    if (n <= 1) {
        isPrime = 0;
    } else {
        for (i = 2; i <= n / 2; ++i) {
            if (n % i == 0) {
                isPrime = 0;
                break;
            }
        }
    }
    if (isPrime) {
        printf("%d is a prime number.\n", n);
    } else {
        printf("%d is not a prime number.\n", n);
    }
    return 0;
}
```

Output:-

```
Enter a number: 45
45 is not a prime number.
```